

Harmonic Resonance Spectra of Biosignals: Separable Harmonicity and Phase-Coupling Axes with a Validated, Swappable-Strategy Toolbox

Methods-paper draft (construct-first narrative). Figures are the composite panels `paper/figures/method_Fig1-9.{png,pdf}` (600 DPI), regenerated by `python -m resonance_paper.paper.make_method_figures` after `python -m resonance_paper.run_all --paper`. All headline numbers are from the paper-grade pass.

Abstract

We present and validate a single- and multi-signal **harmonic resonance** framework that decomposes a time series into a per-frequency **harmonicity** spectrum $H(f)$ (how close the spectral peaks sit to integer ratios), a **phase-coupling** spectrum $PC(f)$ (how stably components hold an $n:m$ phase relation), and their product, a **resonance** spectrum $R(f) = H(f) \cdot PC(f)$, together with a panel of spectral-complexity descriptors and an $n_elec \times n_elec$ connectivity extension. Our central claim is methodological: harmonicity and phase coupling are **separable** axes that merit measurement and reporting as distinct per-frequency spectra, and their product $R = H \cdot PC$ is a principled, interpretable conjunction of the two — *not* a better detector. The implementation is a **strategy registry** of swappable harmonic kernels, ratio kernels, phase estimators, coupling metrics, and combine rules, making the choice of method an explicit, testable variable; we build on established harmonicity and phase-coupling measures and synchronization theory (see Related Work), contributing their unification, validation, and tooling. We establish, in order: **(1)** the two factors are genuinely separable — harmonicity is *exactly* phase-blind ($\rho(H, \text{phase-locking}) = 0.00$), while phase coupling tracks locking; **(2)** the product $R = H \cdot PC$ is the principled conjunction — under independently varied factors a conjunction beats *both* single factors (specificity AUC ≈ 0.99 vs ≤ 0.76) and beats disjunctions, and R adds a phase-coherence requirement that H lacks (phase-scramble: H AUC 0.50 vs R 1.00) — though when the factors co-occur R does not out-*detect* PC , so we present R as an interpretable decomposition, not a superior detector; **(3)** the framework recovers known harmonic structure ($\rho = 0.98$) and $n:m$ coupling (AUC ≈ 1.0); **(4)** we characterize its operating range (detection vs SNR), surrogate-null calibration, and runtime scaling; **(5)** it is competitive with established measures ($n:m$ PLV, harmonic-to-noise ratio); and we demonstrate **(6)** method-choice/mechanism, **(7)** real-biosignal discrimination (matching band power), **(8)** multichannel connectivity, and **(9)** the complexity descriptors as first-class features. We are explicit about scope: the framework adds an interpretable harmonic/coupling axis, and is **not** a replacement for band power on amplitude tasks.

1. Introduction

Biological time series are rich in *harmonic* structure: the QRS complex generates a dense integer-harmonic series; cortical rhythms show cross-frequency relationships and overtones; many physiological oscillators are weakly phase-coupled across frequency. A unifying, interpretable way to quantify *how harmonically organized* a signal is — and at which frequencies — would support comparison across states, modalities, and channels.

We formalize this as a **resonance spectrum**. For a power spectrum $p(f)$, a harmonic kernel $S[i,j]$ scores the musical/harmonic simplicity of each frequency pair and a phase-coupling kernel $\Phi[i,j]$ scores the $n:m$ phase consistency of each pair. PSD-weighted reductions give two per-frequency factors,

$$\begin{aligned} H(f_i) &= p(f_i) \cdot \sum_j S[i,j] \cdot p(f_j) && \text{(harmonicity)} \\ PC(f_i) &= p(f_i) \cdot \sum_j W[i,j] \Phi[i,j] \cdot p(f_j) && \text{(phase coupling)} \end{aligned}$$

and their product $R(f) = H(f) \cdot PC(f)$ is the resonance spectrum. Each spectrum is summarized by complexity descriptors (mean, max, spectral flatness, entropy, spread, Higuchi fractal dimension, peak-harmonic-similarity). The implementation is a strategy registry: harmonic kernels (`harmsim`, `subharm_tension`), ratio kernels (`binary`, `fraction`), phase estimators (`stft`), coupling metrics (`nm_plv`, `nm_plv_canonical`, `nm_pli`, `nm_wpli`, `nm_rrci`), and combine rules (`product`, `geomean`, `harmmean`, `min`, `weighted_log`).

This paper validates the framework construct-first: we begin by establishing that H and PC are *separable* and that the product R is *principled*, then validate recovery, characterize operating range and baselines, and demonstrate applications — being explicit about both what the framework does well and where it does not.

The gap. Existing practice offers no unified, *testable* harmonic axis. Harmonic organization and phase coupling are typically reduced to separate scalar summaries; the method choices that drive them (which harmonic kernel, ratio kernel, phase estimator, coupling metric) are made implicitly and are therefore not comparable across studies; and no toolbox reports per-frequency harmonicity and $n:m$ phase coupling as *dissociable* factors with a calibrated significance null and cross-signal/connectivity extensions. We close this gap and validate the result construct-first.

1.1 Related work

Harmonicity measures. Quantifying "how harmonic is this spectrum?" has a long history in audio/MIR and music perception: the harmonics-to-noise ratio (Boersma 1993), spectral flatness/entropy (Johnston 1988), periodicity/relatedness harmonicity (Stolzenburg 2015), template and spectral-peakiness models (Terhardt et al. 1982; Harrison & Pearce 2020), and harmonic-similarity of interval sets (Gill & Purves 2009); number-theoretic ratio measures (Euler's *gradus*, Tenney height, harmonic entropy) underlie the `harmsim` / `subharm_tension` kernels we reuse from **biotuner** (Bellemare-Pepin & Jerbi 2025). That harmonic organization is a *separable* factor — and a confound for cross-frequency coupling when waveforms are non-sinusoidal — is established by **Dellavale et al. (2020)** (spectral harmonicity / Time-Locked Index on nonlinear oscillators) and Aru et al. (2015); harmonic relationships between EEG peaks are reported by Rodriguez-Larios et al. (2019). Our H is a phase-blind, PSD-weighted, *per-frequency* spectralization of these scalar kernels, distinct from the roughness axis (Plomp & Levelt 1965; Sethares 1993).

Phase coupling, CFC, and inference. n:m phase synchronization originates with Tass et al. (1998), with broadband n:m cross-frequency synchrony shown in cortex by Palva et al. (2005). Our coupling estimators are standard — PLV (Lachaux et al. 1999) and the volume-conduction-robust PLI/wPLI/PPC family (Stam et al. 2007; Vinck et al. 2010, 2011). PC measures phase-**phase** coupling, the rarer complement to the dominant phase-amplitude paradigm (Canolty et al. 2006; Tort et al. 2010). Scheffer-Teixeira & Tort (2016) show that phase-phase coupling claims are easily inflated by weak statistics; our factor-level surrogate-z (AAFT/IAAFT; Theiler et al. 1992; Schreiber & Schmitz 1996) with mandatory aperiodic removal (Donoghue et al. 2020) is a direct response. The swappable-metric + surrogate-z software pattern is shared with **Tensorpac** (Combrisson et al. 2020) and pactools (Dupré la Tour et al. 2017) — but those target phase-*amplitude* coupling, not an n:m phase-phase comodulogram.

Synchronization theory; consonance/FFR. The generative result that ratio complexity sets the n:m locking threshold (Arnold-tongue narrowing) is textbook nonlinear dynamics (Arnold 1965; Pikovsky, Rosenblum & Kurths 2001; Glass & Mackey 1988); the `a_n_a_m` coupling strength is a deliberate special case of weak-coupling phase reduction (Kuramoto 1984; Daido 1996; Ermentrout & Kopell 1991), and ratio complexity is already tied to brainstem lockability by Large's mode-locking program (Lerud et al. 2014; Large et al. 2016). On the auditory side, harmonicity is the load-bearing axis of consonance behaviorally (McDermott et al. 2010; Bowling & Purves 2015) and neurally — Bidelman & Krishnan (2009) and Bidelman (2013) recover the consonance ordering from FFR harmonic salience; intermodulation indexes nonlinear interaction in FFR (Lee et al. 2009) and SSVEP (Boremanse et al. 2013; Norcia et al. 2015). We *reproduce* these established effects as construct validation, not as new findings.

1.2 Contributions

This is an **integration-and-validation** contribution; we are explicit about what is new versus inherited. 1. A **swappable strategy registry** crossing harmonicity kernels with n:m phase-**phase** coupling metrics and combine rules, enabling an empirical kernel-vs-metric comparison — the toolboxes that share this software pattern (Tensorpac, pactools) target phase-*amplitude* coupling, so no off-the-shelf equivalent for n:m phase-phase resonance exists. (*strongest claim*) 2. An explicit, reusable per-frequency $\mathbf{R}(\mathbf{f}) = \mathbf{H}(\mathbf{f}) \cdot \mathbf{PC}(\mathbf{f})$ descriptor that reports harmonicity and phase coupling as two separate spectra and their interpretable conjunction. 3. **Factor-level surrogate-z inference**, with the practical demonstration that testing the composite R can mask a real PC effect, so inference must target the factor that carries it. 4. **Cross-signal and multichannel connectivity** generalizations of the same operator. 5. A **construct-validity bridge**: we recover the classical Arnold-tongue dependence of locking on ratio complexity and use tongue width to validate the harmonicity factor.

The harmonicity construct, the n:m coupling and surrogate machinery, and the synchronization theory are prior art (§1.1); our contribution is their unification, validation, and tooling.

2. Methods

Resonance computation. Single-signal analyses use the `compute_resonance` pipeline (STFT phase estimator; per-frequency factor reduction; product combine) over a 2–45 Hz band at 0.5 Hz resolution unless noted (0.5–45 Hz for ECG). The recommended configuration pairs the `harmsim` harmonic kernel and the `fraction` ratio kernel with the convention-correct `nm_plv_canonical` coupling metric. Cross-signal and connectivity analyses use `compute_cross_resonance` / `compute_cross_resonance_connectivity`.

What the coupling is measured on. Phase coupling is a **phase–phase, polyrhythmic** quantity computed between *pairs of spectral components*: the STFT phase estimator gives the instantaneous phase $\theta(\mathbf{f}, t)$ at each frequency bin, the ratio kernel picks the nearest rational n:m of f_j/f_i , and the coupling metric measures how consistently the generalized phase difference $\mathbf{n} \cdot \boldsymbol{\theta}_i - \mathbf{m} \cdot \boldsymbol{\theta}_j$ stays fixed over time. This is evaluated *within a signal* (cross-frequency coupling between its own peaks → the PC(f) spectrum), *between two signals* (`compute_cross_resonance`), and *across channels* (the connectivity matrix). It is therefore distinct from 1:1 coherence and from phase–amplitude coupling — it detects n:m (polyrhythmic) phase relations between oscillatory components.

Surrogate inference (factor-level). Inference compares an observed spectrum to an ensemble of phase-randomizing surrogates (AAFT for single signals; IAAFT for cross-channel), forming a per-frequency z-score. Crucially, we z-score **each factor (H, PC, R) separately** against the same surrogate ensemble, because R is H-dominated and therefore PSD-driven under a power-preserving null; the phase-coupling factor PC is the correct detector for n:m coupling and must be read as `PC_z`, not in absolute terms.

Aperiodic removal is mandatory across conditions. On pure colored noise, harmonicity rises monotonically with the $1/f$ slope (a confound already noted for spectral harmonicity; Donoghue et al. 2020); we remove the aperiodic component (`remove_aperiodic=True`) whenever conditions differ in background $1/f$. Removal fits a two-parameter power law $P(f) = a \cdot f^{-b}$ to the PSD over the analysis band (non-linear least squares) and subtracts it — a lightweight aperiodic correction, *not* a FOOOF/specparam peak-decomposition.

Signals & data provenance. Synthetic generators with known structure (harmonic ladders, $n:m$ phase locks, polyrhythms, inharmonic/noise controls, independently-varied $H \times PC$) on a matched $1/f$ background (sampling 250–500 Hz, 12–40 s; SNR and seed grids per study). Real EEG is the PhysioNet **EEG Motor Movement/Imagery** database (eegmmidb; Schalk et al. 2004), auto-fetched via `mne.datasets.eegbci` (subjects 1–10): eyes-open vs eyes-closed from the two baseline runs over occipital channels, rest vs motor from a baseline run vs a motor-execution run over central channels, band-pass 1–45 Hz at the native 160 Hz. ECG/PPG/respiration are simulated with `neurokit2` (plus one bundled real ECG clip, tiled to length — noted where used). **Default analysis parameters** (overridable via the registry): 0.5 Hz spectral resolution, 2–45 Hz band (0.5–45 Hz for ECG), `harmsim` harmonic kernel, `fraction` ratio kernel (max-denominator 16), `nm_plv_canonical` coupling metric, `product` combine, aperiodic removal on; STFT overlap is set per study. Detection/discrimination is scored by rank AUC (Mann–Whitney; 0.5 = chance) with percentile bootstrap 95% CIs (2000–3000 resamples, fixed seed); EEG decoding is leave-one-subject-out; surrogate counts are 40–150 per study (more in `--paper`).

Reproducibility note. The suite pins biotuner to the immutable commit `fc158c3` (full SHA in `requirements.txt`); the PyPI 0.4.1 build predates a phase-row alignment fix and silently breaks cross-frequency coupling detection at $f_{min} \geq 1$ Hz. The version string is uninformative (`biotuner.__version__` reads 0.4.1 even on main), so we pin the SHA rather than rely on a version check.

3. Results

3.1 The construct: two separable factors (Fig 1; Study 17)

On a 2-D grid that independently varies ratio complexity (Tenney height) and phase-locking strength κ , **harmonicity is exactly phase-blind**: $\rho(H, \kappa) = 0.00$ — H is flat across locking and ordered only by ratio complexity. Phase coupling tracks locking ($\rho(PC, \kappa) = +0.49$). The one honest asymmetry: PC is **not** ratio-blind ($\rho(PC, \text{complexity}) = -0.47$), because stable mode-locking is physically easier at simple ratios (the Arnold-tongue structure, §3.6). So H and PC are conceptually distinct measurements — cleanly so for H — but not statistically independent in a real oscillator. The construct is also confirmed **generatively** with the physically-correct phase model. We couple harmonically-rich phase oscillators so that an $n:m$ resonance is mediated by the participating harmonics (oscillator-1's n -th meeting oscillator-2's m -th), giving it the leading-order strength a_{n-a_m} (Pikovsky et al.). As coupling K rises each pair locks at its $n:m$ ratio and the framework's PC climbs through the synchronization transition — but the **locking threshold K^* rises with ratio complexity** ($K^* = 3.3, 8.7, 23.5$ for $2:3, 3:4, 4:5$; $\rho(K^*, n-m) = +1.0$): complex ratios, mediated by higher and weaker harmonics, require stronger coupling to lock. This is the Arnold-tongue narrowing (§3.6) seen in the *coupling-strength* dimension. PC tracks this emergent locking directly, whereas the cross-spectral H responds only weakly — and only because locking pulls the frequency onto the exact rational — so the framework's phase coupling, not its harmonicity, reports dynamical synchronization (Fig 1D).

Figure 1 — The construct: harmonicity is phase-blind, phase coupling tracks locking (synthetic grid + generative $n:m$ oscillators)

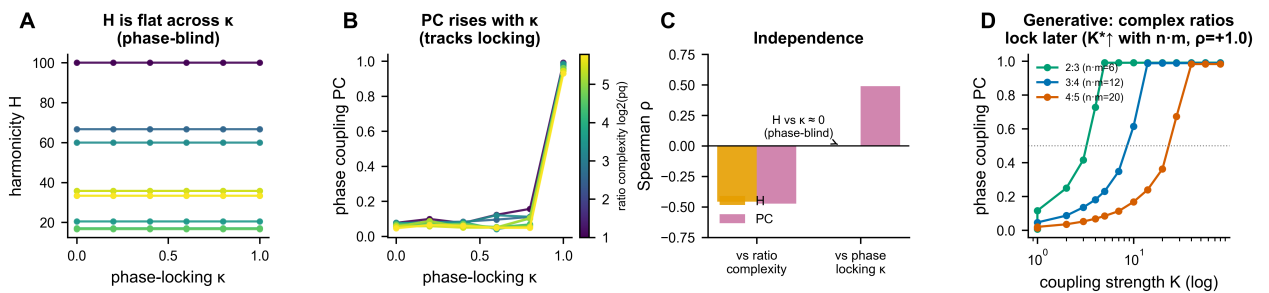


Figure 1. H phase-blind ($\rho(H, \kappa) = 0.00$); PC tracks κ ; independence bars; **generative panel** — harmonically-coupled oscillators, PC locks at the $n:m$ ratio with threshold K^* rising with complexity ($K^* = 3.3/8.7/23.5$ for $2:3/3:4/4:5$, $\rho = +1.0$).

3.2 Why $R = H \cdot PC$ (Fig 2; Study 23; generative panel C from Study 7)

Two results justify the product. **(i) Operator analysis.** With H and PC sampled as *independent* latent factors, a "true resonance" (both high) is separated from single-factor confounds with specificity AUC ≈ 0.99 for conjunctions (product 0.98, geomean 0.98, harmmean 0.99, min 1.00) versus **0.75/0.75 for H/PC alone** and **0.66 for the OR-disjunction (max)** — a conjunction is required, and beats both factors. (`min edges product`; we adopt `product` because $\log R = \log H + \log PC$ gives an additive, interpretable attribution of *why* a resonance is weak.) **(ii) The phase-coherence gate.** Phase-scrambling a harmonic signal preserves its power spectrum exactly, so H is *unchanged* (AUC coherent-vs-scrambled = 0.50, blind) while R collapses (AUC = 1.00): R adds the phase-coherence requirement H lacks. **Honest scope:** when the factors co-occur (coupled oscillators), R does not out-detect PC — its value is the interpretable decomposition, not detection beyond PC . The gate also holds generatively: in the harmonically-coupled $2:3$ oscillators,

R = H-PC stays low while harmonicity is present but the phases are unlocked, and rises only as coupling brings the phases into n:m lock (Fig 2C) — the dynamical analog of the phase-scramble.

Figure 2 — Why R = H-PC: a conjunction (beats both factors; gates harmonicity by phase coherence) — shown abstractly and generatively

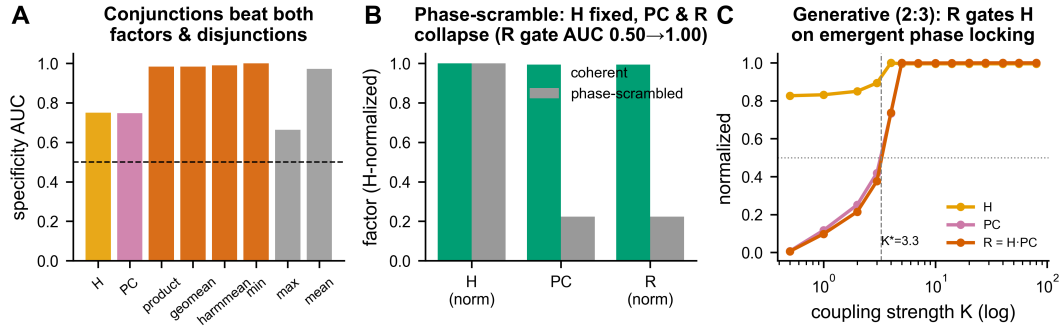


Figure 2. conjunction beats both factors & disjunctions; phase-scramble shows R adds the phase gate H lacks; ****generative panel**** — 2:3 oscillators, R gates harmonicity on the emergent phase locking.

3.3 Ground-truth recovery (Fig 3; Studies 1, 5, 6)

Mean harmonicity increases monotonically with known harmonic richness (Spearman $\rho = 0.98$); harmonic signals separate from inharmonic and from noise at AUC = 1.0. Surrogate-normalized coupling detection recovers known n:m locks: single-signal PC_z AUC = 0.99, phase-coupling matrix entry = 1.0, and cross-signal 1:1/1:2/2:3 locks = 1.0 each; polyrhythm (2:3:4) recovery = 1.0.

Figure 3 — The framework recovers harmonic structure and n:m phase coupling (synthetic ground truth)

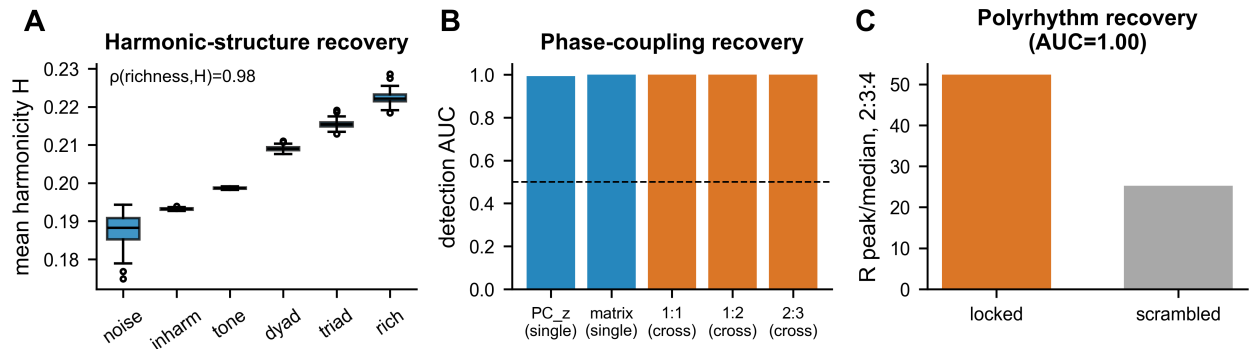


Figure 3. H richness $\rho = 0.98$; n:m coupling AUC ≈ 1.0 ; polyrhythm 1.0.

3.4 Operating characteristics (Fig 4; Study 24)

Detection degrades gracefully with SNR: harmonicity detection holds at AUC ≈ 1.0 down to ≈ -18 dB (0.79 at -24 dB; chance below -30 dB), and n:m coupling has a sharp threshold near -12 dB. The cross PC z-score is **approximately but not perfectly null-calibrated**: on uncoupled pairs the per-instance false-positive rate is ≈ 0.09 at $\alpha = 0.05$ ($n = 120$; null $z = 0.07 \pm 1.07$) — mildly anti-conservative, so a small multiple-surrogate correction (or more surrogates) is advisable for strict type-I control. Runtime is **dominated by the frequency-grid size, not signal length** ($\approx$ flat in length; ≈ 60 ms at 43 bins $\rightarrow 1.1$ s at 172 $\rightarrow 6.6$ s at 430 bins), so cost is budgeted by precision/band, not duration.

Figure 4 — Operating characteristics: SNR robustness, null calibration, scaling

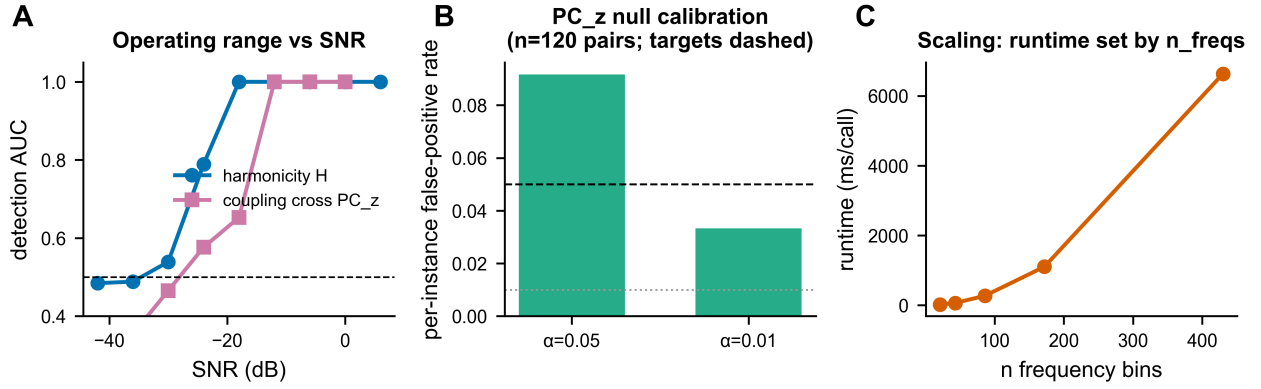


Figure 4. detection AUC vs SNR; PC_z null calibration; runtime vs n_freqs.

3.5 Baselines (Fig 5; Study 25)

Against established measures: framework **PC_z** matches the oracle raw n:m PLV for coupling detection and is **more robust at low SNR** (≈ 6 dB lower detection threshold), while additionally providing a calibrated surrogate null and auto-selecting the n:m ratio. Framework **H** matches the classical harmonic-to-noise ratio at usable SNR (both $AUC \approx 1.0 \geq -18$ dB); honestly, HNR is somewhat more noise-robust at extreme SNR (< -24 dB), while H provides a per-frequency spectrum, handles multi-component/cross-signal cases, and scores inharmonicity explicitly — capabilities HNR's single scalar lacks.

Figure 5 — Baselines: framework factors vs established measures (n:m PLV, HNR)

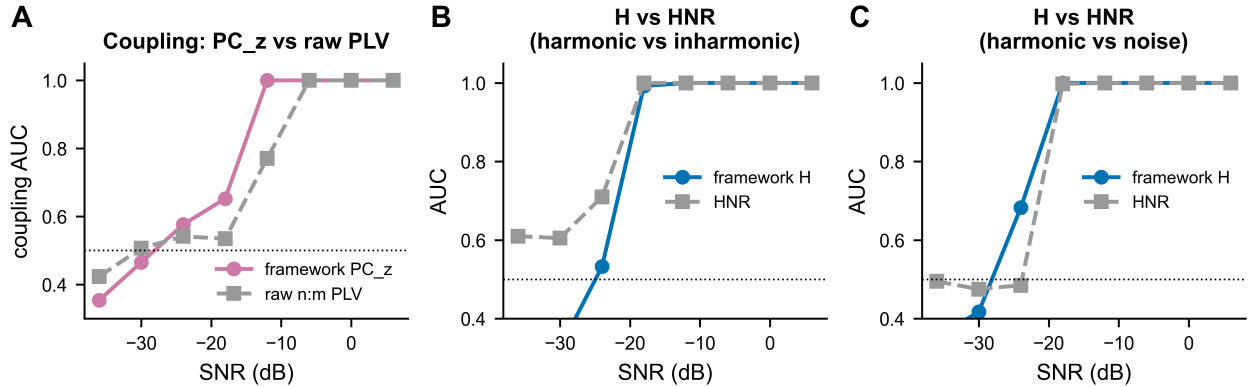


Figure 5. PC_z vs raw n:m PLV; H vs HNR.

3.6 Strategy choice and mechanism (Fig 6; Studies 4, 8)

Sweeping the strategy registry, **detection accuracy saturates** (all 20 kernel×metric configs $\in [0.982, 1.0]$ on the recovery task) — so the practical differentiator is **runtime**: `harmsim` is $\sim 27\times$ faster than `subharm_tension` (0.21 s vs 5.64 s per call) at equal accuracy. Mechanistically, we **recover the classical dependence of lockability on ratio complexity** (Arnold 1965; Pikovsky et al. 2001): in a forced Van der Pol oscillator, Arnold-tongue width correlates with ratio complexity (Spearman $\rho = -0.50$) and with the framework's harmonicity ($\rho = +0.41$) — a sanity-check against established theory that also supplies the physical basis for the H/PC coupling noted in §3.1. (We do not claim this dependence as a finding; it is textbook synchronization theory used here to validate the harmonicity factor.)

Figure 6 — Method choices (strategy registry; accuracy saturates, choose on speed) and the mechanism: harmonic simplicity sets lockability (classical Arnold-tongue)

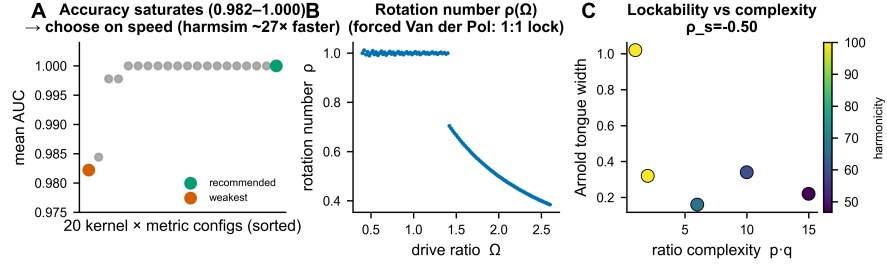


Figure 6. accuracy saturates / harmsim ~27× faster; Arnold-tongue lockability.

3.7 Real biosignals (Fig 7; Studies 2, 3)

On PhysioNet EEG ($n = 10$ subjects, leave-one-subject-out), eyes-open vs eyes-closed decodes at AUC = **0.81**, **matching a relative-alpha-power baseline** (0.81) — an interpretable harmonic-organization axis, not an improvement over band power. The eyes-closed "more harmonic" effect is **1/f-dependent** ($H_{\max}$ AUC 0.83 with aperiodic removal vs 0.29 without). On a rest-run vs motor-run contrast resonance reaches AUC = **0.62** (permutation $p = 0.001$), modestly above the single-band-power baseline (0.54); we flag, however, that this is a *run-level* (baseline-vs-task) comparison confounded by arousal and time-on-task, **not** a clean event-locked motor decode, so we do not read it as a genuine motor advantage. The full feature vector fingerprints **seven signal modalities at 99%** (4-fold CV accuracy 0.99 ± 0.02 ; permutation $p = 0.005$; chance 14%), best read as a multivariate signature on an intentionally easy separation.

Figure 7 — Real biosignals: resonance discriminates states ($\approx$ band power on EO/EC) and fingerprints modality

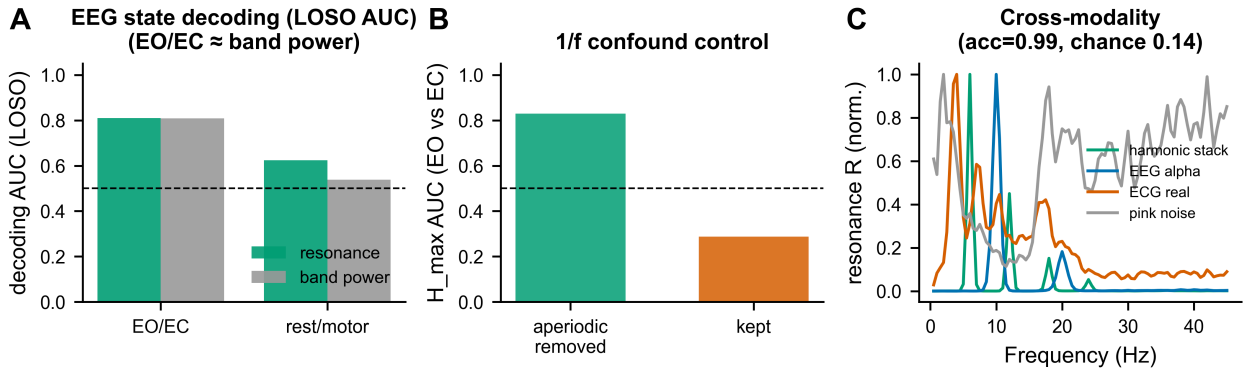


Figure 7. EEG state decoding (EO/EC $\approx$ band power; rest/motor 0.62 > 0.54, run-level confound); 1/f confound control; 7-modality fingerprint (99%).

3.8 The multichannel layer (Fig 8; Study 21)

The cross-resonance connectivity matrix recovers a planted coupled cluster (within-cluster vs rest AUC: H 0.74, PC 0.80, R 0.81). The H matrix is diffuse and PC sharper; $R \approx PC$ because H is near-uniform within the harmonic cluster (consistent with §3.2). **Honest caveat:** the scalar reducer is broadband/overlap-biased, so recovery is driven partly by shared spectra; isolating *pure* cross-channel phase coupling above a PSD-preserving null requires the IAAFT surrogate-z variant (which isolates phase, as H does not survive a power-preserving surrogate).

Figure 8 — Cross-resonance connectivity recovers a planted coupled cluster (overlap-driven; pure phase needs the IAAFT surrogate-z variant, see §3.8)

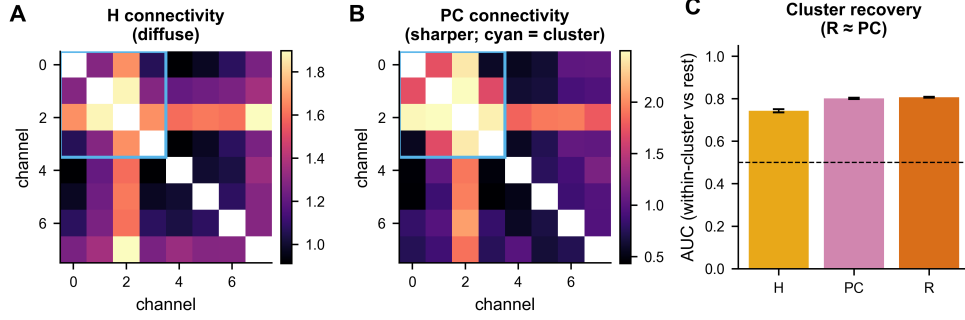


Figure 8. cluster recovery (H diffuse, PC sharp, $R \approx PC$; overlap-driven).

3.9 Spectral-complexity descriptors (Fig 9; Study 22)

The flatness/entropy/spread/Higuchi descriptors of the H/PC/R spectra are first-class features: on signals of known structure they form a distinctive per-class "resonance fingerprint," and they carry information the scalar summaries miss — H_avg conflates pure-tone/inharmonic/narrowband/pink (all ≈ 0.17) while H-spectrum **flatness separates them ($0.17 \rightarrow 0.93$; harmonic-vs-noise AUC = 1.0).**

Figure 9 — Complexity descriptors (flatness/entropy/spread/HFD) of the H/PC/R spectra are first-class features

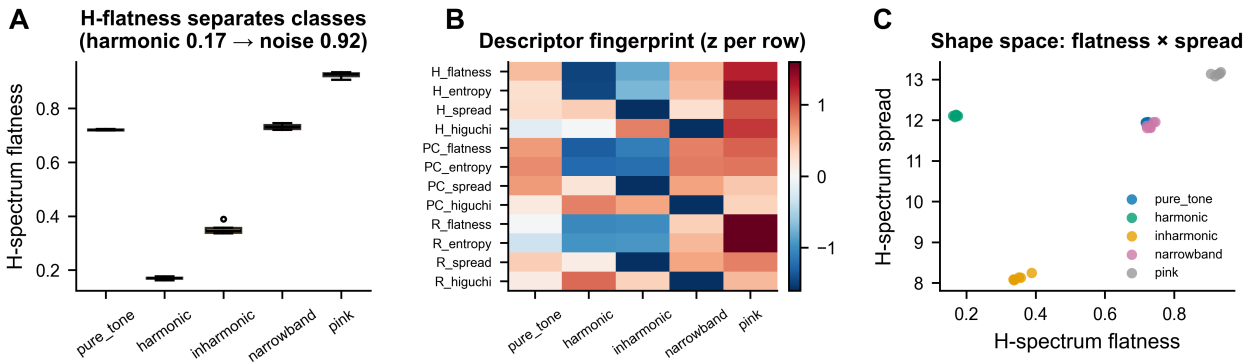


Figure 9. H/PC/R fingerprint; descriptors carry information beyond H_avg.

4. Discussion

The framework is a **principled, separable, honestly-scoped** descriptor of harmonic organization. Its two factors measure genuinely different things — harmonicity is exactly phase-blind — and their product R is the natural conjunction (it beats both factors when they vary independently, and gates harmonicity by phase coherence). The honest boundary is that when harmonicity and locking co-occur (as in real coupled oscillators), R does not out-detect PC; we therefore present R as an interpretable decomposition ($\log R = \log H + \log PC$ attributes a weak resonance to its cause) rather than a superior detector.

Two methodological cautions generalize beyond this toolbox: (i) spectral harmonicity is confounded by the **aperiodic (1/f) component** and must be computed after aperiodic removal whenever a harmonicity *difference* is interpreted; and (ii) surrogate inference must target the **factor that carries the effect** (PC for coupling), not a composite (R) dominated by another factor (H). Scope: the framework adds value for harmonic and cross-frequency-coupling structure, and matches — not beats — band power on single-band amplitude tasks.

5. Limitations and reproducibility

Limitations. (1) *Detection parity.* When harmonicity and phase locking co-occur (real coupled oscillators), $R = H \cdot PC$ does not out-detect PC; R is an interpretable decomposition ($\log R = \log H + \log PC$), not a detector beyond PC (§3.2, §3.8). (2) *Null calibration.* The PC_z surrogate null is mildly anti-conservative (per-instance FPR ≈ 0.09 at $\alpha = 0.05$), so use more surrogates or a multiple-surrogate correction for strict type-I control (§3.4). (3) *Connectivity reducer.* The scalar cross-channel reducer is broadband/overlap-biased; pure cross-channel phase coupling requires the IAAFT surrogate-z variant (§3.8). (4) *Harmonicity under power-preserving nulls.* Because H is fixed by the power spectrum, neither absolute H nor absolute R can be read as harmonic *coupling* beyond what the PSD already implies; only PC_z isolates n:m phase coupling under an AAFT/IAAFT null. (5) *No amplitude advantage.* On single-band amplitude

tasks the framework matches, not beats, band power (§3.7); the rest/motor EEG contrast is additionally a confounded run-level comparison, not a clean motor decode. (6) *Estimator scope*. The phase estimator is STFT-only; Hilbert/wavelet estimators (registry slots present) are a natural next test. (7) *Strength of evidence*. Near-ceiling detection AUCs (≈ 1.0) are on synthetic ground truth and index construct validity, not the magnitude of coupling in real biosignals; the real-EEG effects are modest and reported as such.

Reproducibility. Coupling detection and the connectivity reducer require the alignment-fixed biotuner (pinned commit `fc158c3`, not PyPI 0.4.1; the version string is uninformative). Quick-pass estimates are tightened by `--paper`. One command reproduces every number (`python -m resonance_paper.run_all --paper`) and figure (`python -m resonance_paper.paper.make_method_figures`); see §6.

6. Code & Data Availability

All code, study scripts, and figure builders are in the `biological-resonance` repository (<https://github.com/AntoineBellemare/biological-resonance>). The core engine is **biotuner**, pinned to commit `fc158c3` (`requirements.txt`); PyPI 0.4.1 predates the phase-row alignment fix and silently breaks coupling detection. Real EEG is downloaded on first run from PhysioNet (eegmmidb; Schalk et al. 2004) via MNE; ECG/PPG/respiration are simulated with neurokit2 — no data are redistributed. Reproduce the full numeric suite with `python -m resonance_paper.run_all --paper` and regenerate all figures with `python -m resonance_paper.paper.make_method_figures`.

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Compiled from a literature-positioning pass; verify final bibliographic details (page numbers, DOIs) and the Dellavale et al. attribution against the publisher record before submission.

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